Introduction to CP

From where to start?

- Learn by solving this problems: ICPC Assiut University Community
- Star participate in contests: div4

- By C++ Solve 150- 200 problems of CF: Watermelon Problemset - Codeforces
- Solution: → youtube
- Star participate in contests: div3,4

- STL : Containers and
  - STL Full Series (Bangla) | C++ - YouTube
  - CPS Level - 1 Course (STL Part) [+ paid course?]
- Star participate in contests: div2,3,4

CodeForces

Handle name search: **tourist**
Contest: Div1,2,3,4 [Contests - Codeforces]
Ratings:
Problem Difficulty Level: 800 →
Problem Tags: greedy →

Cf analysis:
CF Explorer
Boro-vai-koise
Codeforces Visualizer
Contest Mania

codeforces solution
friends
tags
difficulty range

Upcoming contest:
CLIST
Codeclock - Coding Calendar - Apps on Google Play

Cf extensions:
CF Analytics - Chrome Web Store
Codeforces Enhancer - Chrome Web Store
Codeforces Friends Tracker
Codeforces Practice Tracker - Chrome Web Store
Codeforces Lite - Dark Mode & Code Runner - Chrome Web Store
Codeforces Helper

Rating range	Title	Division
≥ 4000	eponym	1
3000 — 3999	Legendary Grandmaster	1
2600 — 2999	International Grandmaster	1
2400 — 2599	Grandmaster	1
2300 — 2399	International Master	1
2100 — 2299	Master	1
1900 — 2099	Candidate Master	1/2
1600 — 1899	Expert	2
1400 — 1599	Specialist	2/3
1200 — 1399	Pupil	2/3/4
≤ 1199	Newbie	2/3/4

https://codeforces.com/profile/RakibJoy

Online Contest:	Topic-wise problems:
Codeforces Atcoder Codechef Toph.co	Lightoj Toph.co Uva Cses problem set

Discord Channels:

Bangladesh CP Community

Learning Resources:

The Ultimate Topic List
Algorithms for Competitive Programming
Problem Topics - Codeforces
Camp Resources
শাফায়েতের ব্লগ
bangla-programming-resources: Bangla tutorial, reference and resource list on programming topics
প্রোগ্রামিং প্রবলেম (Programming Problem in Bengali)
Completed Training From Zero
Strivers A2Z DSA Course/Sheet - Crack Any FAANG or PBCs

CF Difficulty Tags	Topic - Wise
800 – 1100	STL, Implementation skill
1200 – 1500	NT, Array Algorithms
1600+	NT, DS, Graph, DP, Math's

Onsite Contest

ACM ICPC: ICPC Asia Dhaka Regional Contest
NCPC: NCPC Onsite Hosted by JU
IUPC: IUPC : National Contests
NGPC: National Girls' Programming Contest
Onsite Contest archive: All National Contests BCS, All Contests toph.co
Onsite Contest slot: 1

==========Code Editor==========

For Competitive Programming

1. sublime text***
2. vscode (xcode for mac book users)**
3. codeblocks*
4. vim

For Development:

1. cursor
2. vscode(xcode for mac book users)
3. devin

Need to Know

Contest strategies: Which problem to solve first in onsite?

Algorithmic skill vs Implementation skill: Which to choose first?

How much problem have to try daily?

How much time to think for a single problem [for easy, medium, hard] ?

Need to Know

Skill vs CGPA: which one?

How efficiently use your time for CP? {Group study, Problem set reading}

For Onsite: Implementation skill or algorithmic skill?

Need to Know

How much problems to solve before shifting into Development?

Frontend vs Backend : Which job?

→ Salary ranges offered by Bangladeshi Software Companies for different positions & Oak Thu

Bangladeshi job market react/php: which one demandable?

→ PHP VS JS [Do you heard about typescript?]

CGPA + Inter mediate CP == ?

Inter mediate CP + Web-development == ?

Good Programmers == ?

Communication skill & Presentation skill: is it important?

Next IUPC and Hackathon

What to do now?

First Step

STL

- Vector
- String
- Pair
- Stack
- Queue
- Deque
- Priority queue
- Set / multiset / unordered_set / unordered_multiset
- map / multimap / unordered_map / unordered_multimap
- Custom comparator
- PDBS / ordered set
- STL algorithms

Bit manipulation

- ✓ Bit wise operators
- ✓ Bit masking
- ✓ STL bitwise functions

Adhoc Basic Concepts

- ✓ Bracket sequence
- ✓ Palindrome
- ✓ Permutation [+ STL]
- ✓ Subsequence
- ✓ Substring
- ✓ sub-array

Array algorithms

- ✓ prefix sum | scanline algorithm
- ✓ difference array
- ✓ array | coordinate compression
- ✓ two pointer
- ✓ sliding window
- ✓ sub-array
- ✓ Kedane's algorithm

Binary search

Hashing 1D, 2D

Number Theory

Next Step : Intermediate → Advanced

- Data-structures
- Graph
- DP
- String

Best of Luck!